

Bayesian hypothesis tests

- Problem formalization
- Introduction to decision theory

Categorical random variables

Provided **classes** $\Omega = \{\omega_m\}_{m=1}^M$, a shorthand like

$$W \sim \text{Categorical}_{\omega_1, \omega_2, \dots, \omega_M} (\pi_{\omega_1}, \pi_{\omega_2}, \dots, \pi_{\omega_M})$$

stands for

- the **values** w that W may attain are $\omega_1, \omega_2, \dots, \omega_M$
- the probability **mass** of W has the form

$$p(w) = \pi_w = \pi_{\omega_1}^{\mathbb{I}_{w=\omega_1}} \pi_{\omega_2}^{\mathbb{I}_{w=\omega_2}} \dots \pi_{\omega_M}^{\mathbb{I}_{w=\omega_M}} = \prod_{\omega \in \Omega} \pi_{\omega}^{\mathbb{I}_{w=\omega}} = \begin{cases} \pi_{\omega_1}, & w = \omega_1 \\ \pi_{\omega_2}, & w = \omega_2 \\ \dots & \\ \pi_{\omega_M}, & w = \omega_M \end{cases} \quad \mathbb{I}_{w=\omega} = \begin{cases} 1, & w = \omega \\ 0, & w \neq \omega \end{cases}$$

- the probability mass of W depends on **parameters**

$$\{\pi_{\omega}\}_{\omega \in \Omega} \subset [0, 1]$$

$$1 = \sum_{\omega \in \Omega} \pi_{\omega}$$

Our classes are **exchangeable**

$$\text{Cat}_{\omega_1, \omega_2, \omega_3} (\pi_{\omega_1}, \pi_{\omega_2}, \pi_{\omega_3}) = \text{Cat}_{\omega_3, \omega_2, \omega_1} (\pi_{\omega_3}, \pi_{\omega_2}, \pi_{\omega_1})$$

Spam detection

In the **spam detection problem**, we seek to determine if an incoming email is spam or ham (legitimate)

For any incoming email, our task is to choose from two **hypothesis**:

- this email is ham
- this email is spam

To decide, we consider:

- Our **baseline**

...ie before reading the message

To date, 20% of our incoming mail had been ham and 80% had been spam

- Our **information**

...ie when reading the message

To date, 1% ham messages contained “urgent” and 99% did not

To date, 85% spam messages contained “urgent” and 15% did not

and evaluate the **odds-statistic**

...ie after reading the message

$$\gamma = \begin{cases} \frac{80\%}{20\%} \frac{85\%}{1\%}, & \text{message contains “urgent”} \\ \frac{80\%}{20\%} \frac{15\%}{99\%}, & \text{message does not contain “urgent”} \end{cases}$$

Our message is classified as **spam** when $\gamma > \gamma_*$ and **ham** when $\gamma \leq \gamma_*$

...choices $\gamma_* = 1, 10, 1000$ designate balanced, safe, and ultra safe settings

Binary hypothesis tests

In a simple hypothesis testing problem, we have two competing hypothesis

- H_0 : null hypothesis
- H_A : alternate hypothesis

and a model of the form

$$\eta \sim \text{Cat}_{H_0, H_A}(\pi_{H_0}, \pi_{H_A})$$
$$w|\eta \sim \begin{cases} \text{Cat}_{\omega_1, \omega_2}(\rho_{H_0 \rightarrow \omega_1}, \rho_{H_0 \rightarrow \omega_2}), & \eta = H_0 \\ \text{Cat}_{\omega_1, \omega_2}(\rho_{H_A \rightarrow \omega_1}, \rho_{H_A \rightarrow \omega_2}), & \eta = H_A \end{cases}$$

This is a Bayesian model and the posterior is

$$p(\eta|w) \propto p(w|\eta)p(\eta) = \rho_{\eta \rightarrow w} \pi_{\eta} = \begin{cases} \rho_{H_0 \rightarrow \omega_1} \pi_{H_0}, & \eta = H_0 \text{ \& } w = \omega_1 \\ \rho_{H_0 \rightarrow \omega_2} \pi_{H_0}, & \eta = H_0 \text{ \& } w = \omega_2 \\ \rho_{H_A \rightarrow \omega_1} \pi_{H_A}, & \eta = H_A \text{ \& } w = \omega_1 \\ \rho_{H_A \rightarrow \omega_2} \pi_{H_A}, & \eta = H_A \text{ \& } w = \omega_2 \end{cases}$$

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Our **odds-statistic** is given by the ratio of the posteriors

$$\gamma(w) = \frac{p(\eta = H_A|w)}{p(\eta = H_0|w)} = \underbrace{\frac{p(w|\eta = H_A)}{p(w|\eta = H_0)}}_{\text{Bayes factor}} \underbrace{\frac{p(\eta = H_A)}{p(\eta = H_0)}}_{\text{prior odds}} = \frac{\rho_{H_A \rightarrow w}}{\rho_{H_0 \rightarrow w}} \frac{\pi_{H_A}}{\pi_{H_0}} = \begin{cases} \frac{\rho_{H_A \rightarrow \omega_1}}{\rho_{H_0 \rightarrow \omega_1}} \frac{\pi_{H_A}}{\pi_{H_0}}, & w = \omega_1 \\ \frac{\rho_{H_A \rightarrow \omega_2}}{\rho_{H_0 \rightarrow \omega_2}} \frac{\pi_{H_A}}{\pi_{H_0}}, & w = \omega_2 \end{cases}$$

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Our **decision** is based on the posterior odds

- We choose H_A , when

$$\gamma(w) > \gamma_* \iff p(\eta = H_A|w) > p(\eta = H_0|w) \gamma_* \qquad \gamma_* = 1$$

- We choose H_0 , when

$$\gamma(w) \leq \gamma_* \iff p(\eta = H_A|w) \leq p(\eta = H_0|w) \gamma_*$$

The **MAP criterion** chooses the hypothesis that has the highest posterior

The **ML criterion** chooses the hypothesis that has the highest likelihood

$$\gamma_* = p(\eta = H_0) / p(\eta = H_A)$$

General hypothesis tests

In a general **hypothesis testing problem**, we have multiple hypothesis $H_1, H_2, \dots, H_M$ and a model of the form

$$\begin{aligned}\eta &\sim \text{Cat}_{H_1, H_2, \dots, H_M}(\pi_{H_1}, \pi_{H_2}, \dots, \pi_{H_M}) \\ w|\eta &\sim L_\eta\end{aligned}$$

Our **decision** is still based on the posterior

The **MAP criterion** chooses the hypothesis that has the highest posterior

$$\begin{aligned}\eta_* &= \operatorname{argmax}_\eta p(\eta|w) \\ &\propto \operatorname{argmax}_\eta p(w|\eta)p(\eta) \\ &= \operatorname{argmax}_\eta L_\eta(w)\pi_\eta\end{aligned}$$

Hypothesis tests with uncertain parameters

A hypothesis testing problem may depend on **uncertain parameters**

$$\begin{aligned}\theta_{H_1} &\sim P_{H_1} \\ \theta_{H_2} &\sim P_{H_2} \\ &\vdots \\ \theta_{H_M} &\sim P_{H_M} \\ \eta &\sim \text{Cat}_{H_1, H_2, \dots, H_M}(\pi_{H_1}, \pi_{H_2}, \dots, \pi_{H_M}) \\ w|\eta, \theta_{H_1}, \theta_{H_2}, \dots, \theta_{H_M} &\sim L_\eta^{\theta_\eta}\end{aligned}$$

Our **decision** is based on the **marginal posterior**

$$\begin{aligned}p(\eta|w) &= \int d\theta_{H_1} \int d\theta_{H_2} \cdots \int d\theta_{H_M} p(\eta, \theta_{H_1}, \theta_{H_2}, \dots, \theta_{H_M}|w) \\ &\propto \int d\theta_{H_1} \int d\theta_{H_2} \cdots \int d\theta_{H_M} p(w|\eta, \theta_{H_1}, \theta_{H_2}, \dots, \theta_{H_M}) p(\eta, \theta_{H_1}, \theta_{H_2}, \dots, \theta_{H_M}) \\ &= \int d\theta_{H_1} \int d\theta_{H_2} \cdots \int d\theta_{H_M} p(w|\eta, \theta_{H_1}, \theta_{H_2}, \dots, \theta_{H_M}) p(\eta) p(\theta_{H_1}) p(\theta_{H_2}) \cdots p(\theta_{H_M})\end{aligned}$$

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$$\theta_{H_1} \sim P_{H_1}$$

$$\theta_{H_2} \sim P_{H_2}$$

$$\vdots$$

$$\theta_{H_M} \sim P_{H_M}$$

$$\eta \sim \text{Cat}_{H_1, H_2, \dots, H_M}(\pi_{H_1}, \pi_{H_2}, \dots, \pi_{H_M})$$

$$w|\eta, \theta_{H_1}, \theta_{H_2}, \dots, \theta_{H_M} \sim L_{\eta}^{\theta_{\eta}}$$

Our **decision** is based on the **marginal posterior**

$$\begin{aligned} p(\eta|w) &\propto \int d\theta_{H_1} \int d\theta_{H_2} \cdots \int d\theta_{H_M} p(w|\eta, \theta_{H_1}, \theta_{H_2}, \dots, \theta_{H_M}) p(\eta) p(\theta_{H_1}) p(\theta_{H_2}) \cdots p(\theta_{H_M}) \\ &= p(\eta) \int d\theta_{H_1} \int d\theta_{H_2} \cdots \int d\theta_{H_M} p(w|\eta, \theta_{\eta}) p(\theta_{H_1}) p(\theta_{H_2}) \cdots p(\theta_{H_M}) \\ &= p(\eta) \int d\theta_{\eta} p(w|\eta, \theta_{\eta}) p(\theta_{\eta}) \end{aligned}$$

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$$\vdots$$

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$$w|\eta, \theta_{H_1}, \theta_{H_2}, \dots, \theta_{H_M} \sim L_{\eta}^{\theta_{\eta}}$$

Our **decision** is based on the **marginal posterior**

$$p(\eta|w) \propto p(\eta) \int d\theta_{\eta} p(w|\eta, \theta_{\eta}) p(\theta_{\eta}) = \pi_{\eta} \int d\theta_{\eta} L_{\eta}^{\theta_{\eta}}(w) P_{\eta}(\theta_{\eta})$$

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$$\theta_{H_2} \sim P_{H_2}$$

$$\vdots$$

$$\theta_{H_M} \sim P_{H_M}$$

$$\eta \sim \text{Cat}_{H_1, H_2, \dots, H_M}(\pi_{H_1}, \pi_{H_2}, \dots, \pi_{H_M})$$

$$w|\eta, \theta_{H_1}, \theta_{H_2}, \dots, \theta_{H_M} \sim L_{\eta}^{\theta_{\eta}}$$

Our **decision** is based on the **marginal posterior**, which requires evaluation of the **marginal likelihood** integrals

$$\ell_{H_1} = \int d\theta_{H_1} L_{H_1}^{\theta_{H_1}}(w) P_{H_1}(\theta_{H_1})$$

$$\ell_{H_2} = \int d\theta_{H_2} L_{H_2}^{\theta_{H_2}}(w) P_{H_2}(\theta_{H_2})$$

$$\dots$$